

# Kristin Branson

SENIOR GROUP LEADER

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📍 kristinbranson | 📞 Kristin Branson | 🐦 @kristinmbranson

## Summary

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I develop new and impactful ways to use computer vision (CV) and machine learning (ML) to gain insight into scientific questions. I do this by 1) finding new scientific questions that can be framed as CV&ML problems, 2) engineering practical and integrative solutions, and 3) making these systems usable by others so that these methods are broadly adopted and applied to new problems. I was a pioneer of video-based analysis of animal behavior, and my work helped establish this as an important and ubiquitous technique in neuroscience and biology more generally.

## Education

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### University of California, San Diego

PH.D., M.S., COMPUTER SCIENCE

*La Jolla, CA*

2007, 2002

Dissertation Title: *Tracking Multiple Mice through Severe Occlusions*

Advisers: Serge Belongie and Sanjoy Dasgupta

### Harvard University

A.B. *Cum Laude*, COMPUTER SCIENCE

*Cambridge, MA*

2000

## Experience

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### HHMI Janelia Research Campus

SENIOR GROUP LEADER

HEAD OF COMPUTATION & THEORY

GROUP LEADER

*Ashburn, VA*

2017-present

2017-2022

2010-2017

### California Institute of Technology

POSTDOCTORAL RESEARCHER

Advisers: Pietro Perona and Michael Dickinson

*Pasadena, CA*

2007-2009

### NASA Intern

GRADUATE STUDENT RESEARCHER

Research Institute for Advanced Computer Science

UNDERGRADUATE RESEARCHER

NASA Academy Program

*NASA Ames RC*

Summer 2000 & 2001

*NASA Armstrong Flight RC*

Summer 1999

## Open-Source Software

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I lead, co-developed, and co-maintain the following open-source pieces of software:

- APT: The Animal Part Tracker, <http://kristinbranson.github.io/APT/>
- BABAM: The Browseable Atlas of Behavior-Anatomy Maps, <https://kristinbranson.github.io/BABAM/>
- BIAS: Basic Image Acquisition Software <http://stuff.iorodeo.com/notes/bias/>
- JAABA: Janelia Automatic Animal Behavior Annotator, <http://jaaba.sourceforge.net/>
- Ctrax: The Caltech Multiple Walking Fly Tracker, <http://ctrax.sourceforge.net/>

## Conferences Organized

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- Multi-Agent Behavior Workshop, CVPR, June, 2022.
- 4D Cellular Physiology Reimagined: Theory as a Principal Component, September, 2021.
- Janelia Conference: Women in Computational Biology, November 2019.
- Bioimage Computing Workshop, CVPR, June 2019.
- Society for Neuroscience Virtual Conference on Machine Learning, June 2019.
- Janelia Junior Scientist Workshop on Machine Learning and Computer Vision, April 2019, October 2017, October 2016, October 2015.
- Janelia Conference: What Can Machine Learning Do for Neuroscience and Vice-Versa?, November 2010.

## Honors and Awards

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- Selected as one of Cell's "40 under 40" scientists in commemoration of their 40th anniversary, 2015.
- Faculty of 1000 Recommendation for JAABA, 2013.
- NASA Graduate Student Researcher Program Fellowship, 2003-2006.

## Scientific Leadership

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- Led a lab from 2010-present.
- Established the Computation and Theory Research Area at Janelia (10 labs) and served as inaugural Head from 2017-2022. Set the overall vision and direction for computational research at Janelia; recruited, made hiring and renewal decisions, and set budgets for Group Leaders; initiated development of Janelia's GPU cluster; initiated Janelia's Theory Fellow program.
- As part of Senior Leadership at Janelia, participated in institute-wide planning and decision-making, 2017-2022.
- Co-created the Janelia Diversity, Equity, and Inclusion Committee, 2018. Chaired committee 2018-2020. Participated 2020-2022. As part of this committee, we have researched and proposed initiatives to Janelia leadership, invited speakers to present at Janelia, and started a DEI reading group.
- Started, co-organized, and contributed to Computer Vision and Machine Learning weekly reading group and internal seminar series at Janelia, 2014-present.
- Started and co-organized the Janelia Computation and Theory Seminar series, bi-weekly, 2018-present.
- Participated in and led a Women and Non-binary Scientist Mentoring Group at Janelia, 2018-present.
- Started and co-organized Janelia Computation and Theory social event to build community at Janelia, 2018-2020.

## Outreach

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- Directed the Cajal Machine Learning for Neuroscience Summer School (Summer, 2023).
- Taught classes on quantitative analysis of fly behavior at the Marine Biology Lab (June-July, 2022) and Howard University, an HBCU (September-November, 2022).
- Started and organized "Hour of Code" bi-monthly outreach activity for Janelia scientists to share their enthusiasm for and the impact of programming with grade-school children at local elementary schools and libraries, 2017-2020. This was the first student outreach activity at Janelia, and instigated Janelia to build a Community Relations group to support future efforts.
- Worked with high school teachers at Loudoun Academies of Science to develop a machine learning course for high school students, 2018. Volunteer to provide project mentorship to students in these classes, 2018-present.
- Participated in "Raising Excitement for Science, Engineering, and Technology" (RESET) Education Outreach Program, 2018-2020, program organized by lab member Alice Robie.
- Taught/mentored in several summer and winter schools, including Neuromatch, FENS, Cajal, Cold Spring Harbor, Jackson Laboratory, and FLIACT. Example of a Colab notebook developed for this purpose: <https://bit.ly/jaxpose>.

## Publications

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1. K. A. Horstmann, E. Lin, A. A. Robie, J. J. Sun, **K. Branson**. “A case study of evaluating AI agents on a neuroscience data-to-discovery pipeline.” *arXiv* 2026. [arXiv:2606.07718](#)
2. L.-Q. Zhang, **K. Branson**. “Neurodata Without Boredom: Benchmarking Agentic AI for Data Reuse.” *arXiv* 2026. [arXiv:2605.12808](#)
3. E. Eyjólfssdóttir, **K. Branson**. “Agent-Centric Animal Pose Forecasting.” *In review*, 2026.
4. A. Sarkar, E. Lin, Z. Liu, **K. Branson**, J. J. Sun. “AgentSSL: Can MLE Agents Leverage Unlabeled Data?” *In review*, 2026.
5. G. M. Rubin, C. Managan, M. Dreher, E. Kim, S. Miller, K. Boone, A. Robie, A. L. Taylor, **K. Branson**, C. E. Schretter, A. G. Otopalik. “Networks of sexually dimorphic neurons that regulate social behaviors in *Drosophila*.” *bioRxiv* 2025. [doi:10.1101/2025.10.21.683766](#)
6. H. Chiu, A. A. Robie, **K. Branson**, T. Vipppa, S. Epstein, G. M. Rubin, D. J. Anderson, C. E. Schretter. “Cell type-specific contributions to a persistent aggressive internal state in female *Drosophila*.” *eLife*, 2025. [doi:10.7554/eLife.88598.3](#)
7. J. A. Keller, I. S. Kwak, A. K. Stark, M. Pachitariu, **K. Branson**, J. T. Dudman. “Cortical control of innate behavior from subcortical demonstration.” *bioRxiv*, 2025. [doi:10.1101/2025.02.12.637930](#)
8. R. Vaxenburg, I. Siwanowicz, J. Merel, A. A. Robie, C. Morrow, G. Novati, Z. Stefanidi, G. M. Card, M. B. Reiser, M. M. Botvinick, **K. Branson**, Y. Tassa, S. C. Turaga. “Whole-body simulation of realistic fruit fly locomotion with deep reinforcement learning.” *Nature*, vol. 643, pp. 1312–1320, 2025. [doi:10.1038/s41586-025-09029-4](#)
9. A. A. Robie, A. L. Taylor, C. E. Schretter, M. Kabra, **K. Branson**. “The Fly Disco: Hardware and software for optogenetics and fine-grained fly behavior analysis.” *bioRxiv*, 2024. [doi:10.1101/2024.11.04.621948](#)
10. C. E. Schretter, T. H. Sten, N. Klapoetke, M. Shao, A. Nern, M. Dreher, D. Bushey, A. A. Robie, A. L. Taylor, **K. Branson**, A. Otopalik, V. Ruta, G. M. Rubin. “Social state alters vision using three circuit mechanisms in *Drosophila*.” *Nature*, vol. 637, pp. 646–653, 2024. [doi:10.1038/s41586-024-08255-6](#)
11. B. Gorko, I. Siwanowicz, K. Close, C. Christoforou, K. L. Hibbard, M. Kabra, ... **K. Branson**, G. Ihrke, S. J. Huston. “Motor neurons generate pose-targeted movements via proprioceptive sculpting.” *Nature*, vol. 628, no. 8008, pp. 596–603, 2024. [doi:10.1038/s41586-024-07222-5](#)
12. M. Isaacson, J. Eliason, A. Nern, E. Rogers, G. Lott, T. Tabachnik, W. Rowell, A. Edwards, W. Korff, G. Rubin, **K. Branson**, M. B. Reiser. “Small-field visual projection neurons detect translational optic flow and support walking control.” *bioRxiv* 2023.06.21.546024, 2023. [doi:10.1101/2023.06.21.546024](#)
13. J. J. Sun, A. Ulmer, D. Chakraborty, B. Geuther, E. Hayes, H. Jia, V. Kumar, Z. Partridge, A. Robie, C. E. Schretter, C. Sun, K. Sheppard, P. Uttarwar, P. Perona, Y. Yue, **K. Branson**, A. Kennedy. “The MABe22 Benchmarks for Representation Learning of Multi-Agent Behavior.” *ICML*, 2023. [doi:10.48550/arXiv.2207.10553](#)
14. J. Z. Guo, B. Sauerbrei, J. Cohen, M. Mischiatti, A. Graves, **K. Branson**, A. Hantman. “Disrupting cortico-cerebellar communication impairs dexterity.” *eLife*, vol. 10, p. e65906, 2021. [doi:10.7554/eLife.65906](#)
15. C. E. Schretter, Y. Aso, M. Dreher, A. A. Robie, M.-J. Dolan, N. Chen, M. Ito, T. Yang, R. Parekh, **K. Branson**, G. M. Rubin. “Cell types and neuronal circuitry underlying female aggression in *Drosophila*.” *eLife*, vol. 9, p. e58942, 2020. [doi:10.7554/eLife.58942](#)
16. D. J. Im, I. S. Kwak, **K. Branson**. “Evaluation metrics for behavior modeling.” *arXiv* 2007.12298, 2020. [doi:10.48550/arXiv.2007.12298](#)
17. I. S. Kwak, D. Kriegman, **K. Branson**. “Detecting the starting frame of actions in video.” *WACV*, 2020. [doi:10.1109/WACV45572.2020.9093405](#)
18. B. Sauerbrei, J.-Z. Guo, M. Mischiatti, W. Guo, M. Kabra, N. Verma, B. Mensch, **K. Branson**, A. Hantman. “Cortical pattern generation during dexterous movements is input-driven.” *Nature*, vol. 577, pp. 386–391, 2019. [doi:10.1038/s41586-019-1869-9](#)
19. D. J. Im, S. Prakhya, J. Yan, S. Turaga, **K. Branson**. “Importance weighted adversarial variational autoencoders for spike inference from calcium imaging data.” *arXiv* 1906.03214, 2019. [doi:10.48550/arXiv.1906.03214](#)
20. J. M. Ache, S. Namiki, A. Lee, **K. Branson**, G. M. Card. “State-dependent decoupling of sensory and motor circuits underlies behavioral flexibility in *Drosophila*” *Nature Neuroscience*, vol. 22, no. 7, pp. 1123–1131, s41593-019-0413-4, 2019. [doi:10.1038/s41593-019-0413-4](#)

21. D. J. Im, N. Verma, **K. Branson**. “Stochastic neighbor embedding under f-divergences.” *arXiv* 1811.01247, 2018. [doi:10.48550/arXiv.1811.01247](https://doi.org/10.48550/arXiv.1811.01247)
22. D. J. Im, H. Ma, G. W. Taylor, **K. Branson**. “Quantitatively evaluating GANs with divergences proposed for training.” *ICLR*, 2018. [10.48550/arXiv.1803.01045](https://doi.org/10.48550/arXiv.1803.01045)
23. K. McDole, L. Guignard, F. Amat, A. Berger, G. Malandain, L. A. Royer, S. C. Turaga, **K. Branson**, P. J. Keller. “In toto imaging and reconstruction of post-implantation mouse development at the single-cell level.” *Cell*, vol. 175, no. 3, pp. 859–876, 2018. [doi:10.1016/j.cell.2018.09.031](https://doi.org/10.1016/j.cell.2018.09.031)
24. I. F. Rodriguez, R. Megret, R. Egnor, **K. Branson**, J. L. Agosto, T. Giray, E. Acuna. “Multiple animals tracking in video using part affinity fields.” *Visual Observation and Analysis of Vertebrate And Insect Behavior*, 2018. [\[link\]](#)
25. I. F. Rodriguez, **K. Branson**, E. Acuna, J. L. Agosto-Rivera, T. Giray, R. Megret. “Honeybee detection and pose estimation using convolutional neural networks.” *Reconnaissance des Formes, Image, Apprentissage et Perception*, 2018. [\[link\]](#)
26. **K. Branson**. “A deep (learning) dive into a cell.” *Nature Methods*, vol. 15, no. 4, p. 253, 2018. [doi:10.1038/nmeth.4658](https://doi.org/10.1038/nmeth.4658)
27. A. A. Robie, J. Hirokawa, A. W. Edwards, L. A. Umayam, A. Lee, M. L. Phillips, G. M. Card, W. Korff, G. M. Rubin, J. H. Simpson, M. B. Reiser, **K. Branson**. “Mapping the neural substrates of behavior.” *Cell*, vol. 170, no. 2, pp. 393–406, 2017. [doi:10.1016/j.cell.2017.06.032](https://doi.org/10.1016/j.cell.2017.06.032)
28. R. Sen, M. Wu, **K. Branson**, A. Robie, G. M. Rubin, B. J. Dickson. “Moonwalker descending neurons mediate visually evoked retreat in *Drosophila*.” *Current Biology*, vol. 27, no. 5, pp. 766–771, 2017. [doi:10.1016/j.cub.2017.02.008](https://doi.org/10.1016/j.cub.2017.02.008)
29. A. A. Robie, K. M. Seagraves, S. R. Egnor, **K. Branson**. “Machine vision methods for analyzing social interactions.” *Journal of Experimental Biology*, vol. 220, no. 1, pp. 25–34, 2017. [doi:10.1242/jeb.142281](https://doi.org/10.1242/jeb.142281)
30. D. J. Im, M. Tao, **K. Branson**. “An empirical analysis of deep network loss surfaces.” *arXiv* 1612.04010, 2016. [doi:10.48550/arXiv.1612.04010](https://doi.org/10.48550/arXiv.1612.04010)
31. E. Eyjolfsson, **K. Branson**, Y. Yue, P. Perona. “Learning recurrent representations for hierarchical behavior modeling.” *International Conference on Learning Representations*, 2017. [doi:10.48550/arXiv.1611.00094](https://doi.org/10.48550/arXiv.1611.00094)
32. S. R. Egnor, **K. Branson**. “Computational analysis of behavior.” *Annual Review of Neuroscience*, vol. 39, pp. 217–236, 2016. [doi:10.1146/annurev-neuro-070815-013845](https://doi.org/10.1146/annurev-neuro-070815-013845)
33. J.-Z. Guo, A. R. Graves, W. W. Guo, J. Zheng, A. Lee, J. Rodriguez-Gonzalez, N. Li, J. J. Macklin, J. W. Phillips, B. D. Mensh, **K. Branson**, A. Hantman. “Cortex commands the performance of skilled movement.” *eLife*, vol. 4, p. e10774, 2015. [doi:10.7554/eLife.10774](https://doi.org/10.7554/eLife.10774)
34. **K. Branson**, J. Freeman. “Imaging the neural basis of locomotion.” *Cell*, vol. 163, no. 3, pp. 541–542, 2015. [10.1016/j.cell.2015.10.014](https://doi.org/10.1016/j.cell.2015.10.014)
35. W. C. Lemon, S. R. Pulver, B. Höckendorf, K. McDole, **K. Branson**, J. Freeman, P. J. Keller. “Whole-central nervous system functional imaging in larval *Drosophila*.” *Nature Communications*, vol. 6, p. 7924, 2015. [doi:10.1038/ncomms8924](https://doi.org/10.1038/ncomms8924)
36. T. Ohyama, C. M. Schneider-Mizell, R. D. Fetter, J. V. Aleman, R. Franconville, M. Rivera-Alba, B. D. Mensh, K. M. Branson, J. H. Simpson, J. W. Truman, A. Cardona, M. Zlatić. “A multilevel multimodal circuit enhances action selection in *Drosophila*.” *Nature*, vol. 520, no. 7549, pp. 633–639, 2015. [doi:10.1038/nature14297](https://doi.org/10.1038/nature14297)
37. N. Verma, **K. Branson**. “Sample complexity of learning mahalanobis distance metrics.” *NIPS*, pp. 2584–2592, 2015. [doi:10.48550/arXiv.1505.02729](https://doi.org/10.48550/arXiv.1505.02729)
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39. **K. Branson**. “Distinguishing seemingly indistinguishable animals with computer vision.” *Nature Methods*, vol. 11, no. 7, pp. 721–723, 2014. [doi:10.1038/nmeth.3004](https://doi.org/10.1038/nmeth.3004)
40. A. I. Dell, J. A. Bender, **K. Branson**, I. D. Couzin, G. G. de Polavieja, L. P. Noldus, A. Pérez-Escudero, P. Perona, A. D. Straw, M. Wikelski, U. Brose. “Automated image-based tracking and its application in ecology.” *Trends in Ecology & Evolution*, vol. 29, no. 7, pp. 417–428, 2014. [doi:10.1016/j.tree.2014.05.004](https://doi.org/10.1016/j.tree.2014.05.004)
41. Y. Aso, D. Sitaraman, T. Ichinose, K. R. Kaun, K. Vogt, G. Belliart-Guérin, P.-Y. Plaçais, A. A. Robie, N. Yamagata, C. Schnaitmann, W. J. Rowell, R. M. Johnston, T.-T. B. Ngo, N. Chen, W. Korff, M. N. Nitabach, U. Heberlein, T. Preat, **K. Branson**, H. Tanimoto, G. M. Rubin. “Mushroom body output neurons encode valence and guide memory-

based action selection in *Drosophila*.” *eLife*, vol. 3, p. e04580, 2014. [doi:10.7554/eLife.04580](https://doi.org/10.7554/eLife.04580)

42. F. Amat, W. Lemon, D. P. Mossing, K. McDole, Y. Wan, **K. Branson**, E. W. Myers, P. J. Keller. “Fast, accurate reconstruction of cell lineages from large-scale fluorescence microscopy data.” *Nature Methods*, vol. 11, no. 9, pp. 951–958, 2014. [doi:10.1038/nmeth.3036](https://doi.org/10.1038/nmeth.3036)
43. D. G. Tervo, M. Proskurin, M. Manakov, M. Kabra, A. Vollmer, **K. Branson**, A. Y. Karpova. “Behavioral variability through stochastic choice and its gating by anterior cingulate cortex.” *Cell*, vol. 159, no. 1, pp. 21–32, 2014. [doi:10.1016/j.cell.2014.08.037](https://doi.org/10.1016/j.cell.2014.08.037)
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45. F. Zabala, P. Polidoro, A. Robie, **K. Branson**, P. Perona, M. H. Dickinson. “A simple strategy for detecting moving objects during locomotion revealed by animal-robot interactions.” *Current Biology*, vol. 22, no. 14, pp. 1344–1350, 2012. [doi:10.1016/j.cub.2012.05.024](https://doi.org/10.1016/j.cub.2012.05.024)
46. A. D. Straw, **K. Branson**, T. R. Neumann, M. H. Dickinson. “Multi-camera real-time three-dimensional tracking of multiple flying animals.” *Journal of The Royal Society Interface*, vol. 8, no. 56, pp. 395–409, 2010. [doi:10.1098/rsif.2010.0230](https://doi.org/10.1098/rsif.2010.0230)
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48. S. Agarwal, **K. Branson**, S. Belongie. “Higher order learning with graphs.” *ICML*, pp. 17–24, 2006. [doi:10.1145/1143844.1143847](https://doi.org/10.1145/1143844.1143847)
49. **K. Branson**, S. Belongie. “Tracking multiple mouse contours (without too many samples).” *CVPR*, vol. 1, pp. 1039–1046, 2005. [doi:10.1109/CVPR.2005.349](https://doi.org/10.1109/CVPR.2005.349)
50. **K. Branson**, V. Rabaud, S. Belongie. “Three brown mice: See how they run.” *VS-PETS Workshop at ICCV*, 2003. [\[link\]](#)
51. G. W. Cottrell, K. M. Branson, A. J. Calder. “Do expression and identity need separate representations?” *Proceedings of the Annual Meeting of the Cognitive Science Society*, vol. 24, no. 24, 2002. [\[link\]](#)

## Mentorship

| Name                    | Years in lab | Degree received              | Current Position                                                                                                  |
|-------------------------|--------------|------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Alice Robie             | 2010-present |                              | Senior Scientist in my lab                                                                                        |
| Mayank Kabra            | 2011-2013    |                              | Machine learning consultant, Kabra Consulting                                                                     |
| Marta Rivera-Alba       | 2013-2016    |                              | Head of Data Science at Causal Foundry                                                                            |
| Nakul Verma             | 2013-2017    |                              | Professor, Computer Science, Columbia U.                                                                          |
| Kelly Seagraves         | 2015-2016    |                              | Senior Advisor in the U.S. Department of State's Office of the Special Envoy for Critical and Emerging Technology |
| Roian Egnor             | 2015-present |                              | Senior Scientist in my lab                                                                                        |
| Iljung (Sam) Kwak       | 2015-2023    | PhD, CS, UCSD, 2019          | 3D Machine Learning Engineer at Nuwa                                                                              |
| Jiwoong (Daniel) Im     | 2016-2020    |                              | PhD student, CS, NYU                                                                                              |
| Rutuja Patil            | 2017-2024    |                              | Software Engineer in Edge Computing                                                                               |
| Ivan (Felipé) Rodriguez | 2017-2018    | MS, CS, U. Puerto Rico, 2019 | PhD student, Cognitive Science, Brown                                                                             |
| Lingqi Zhang            | 2023-present |                              | Theory Fellow with my lab                                                                                         |
| Aniket Ravan            | 2024-present |                              | Machine Learning Researcher in my lab                                                                             |
| Eyrun Eyjolfssdottir    | 2024-present |                              | Machine learning consultant                                                                                       |
| Kai Horstmann           | 2025-present |                              | PhD student, CS, Cornell                                                                                          |
| Yijia Dai               | 2025-present |                              | PhD student, CS, Cornell                                                                                          |

With members of my lab, I’ve also co-mentored two undergraduate students and three high school students. 2/3 of my mentees have been from groups historically marginalized in STEM. As Head of Computation and Theory, I provided mentorship and advice to the other 9 Group Leaders within this Research Area.

## Outside the lab

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- I love rock climbing, hiking, and generally being outside in the sun. My favorite is brainstorming zany machine learning and science ideas while doing these activities with other scientists.
- I enjoy word puzzles, particularly crossword puzzles.